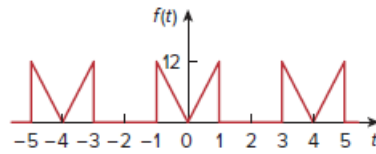


Problem

Calculate the Fourier coefficients for the function in Fig. 17.60.

Figure 17.60:



Step-by-step solution

Step 1 of 5

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} 12t, & 0 < t < 1 \\ 0, & 1 < t < 3 \\ -12(t-4), & 3 < t < 4 \end{cases}$$

Step 2 of 5

The time period is, $T = 4$.

Determine the value of ω_0 .

$$\begin{aligned}\omega_0 &= \frac{2\pi}{4} \\ &= \frac{\pi}{2}\end{aligned}$$

Determine the value of a_0 .

$$\begin{aligned}a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{4} \int_0^4 f(t) dt \\ &= \frac{1}{4} \int_0^1 f(t) dt + \frac{1}{4} \int_1^3 f(t) dt + \frac{1}{4} \int_3^4 f(t) dt \\ &= \frac{1}{4} \int_0^1 [12t] dt + \frac{1}{4} \int_1^3 0 \cdot dt + \frac{1}{4} \int_3^4 (-12)(t-4) dt \\ &= 3 \left[\frac{t^2}{2} \right]_{t=0}^{t=1} + 0 - 3 \left[\frac{(t-4)^2}{2} \right]_{t=3}^{t=4} \\ &= 3 \left[\frac{1}{2} - 0 \right] - 3 \left[\frac{(4-4)^2}{2} - \frac{(3-4)^2}{2} \right] \\ &= 3 \left(\frac{1}{2} \right) - 3 \left[0 - \frac{1}{2} \right] \\ &= 3\end{aligned}$$

Step 3 of 5

Determine the value of a_n .

$$\begin{aligned}a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\ &= \frac{2}{4} \int_0^4 f(t) \cos(n\omega_0 t) dt \\ &= \frac{1}{2} \int_0^1 f(t) \cos(n\omega_0 t) dt + \frac{1}{2} \int_1^3 f(t) \cos(n\omega_0 t) dt + \frac{1}{2} \int_3^4 f(t) \cos(n\omega_0 t) dt \\ &= \frac{1}{2} \int_0^1 [12t] \cos\left(\frac{n\pi}{2} t\right) dt + \frac{1}{2} \int_1^3 0 \cdot \cos\left(\frac{n\pi}{2} t\right) dt + \frac{1}{2} \int_3^4 (-12)(t-4) \cos\left(\frac{n\pi}{2} t\right) dt\end{aligned}$$

Step 4 of 5

Use the following formula:

$$\int uv = u \int v dx - \int u' \left(\int v dx \right) dx$$

Simplify the equation further,

$$\begin{aligned} a_n &= \left\{ \frac{12}{n\pi} \left[t \sin \left(\frac{n\pi}{2} t \right) \right]_{t=0}^{t=1} + \frac{24}{n^2 \pi^2} \cos \left(\frac{n\pi}{2} t \right) \Big|_{t=0}^{t=1} + 0 + \frac{48}{n\pi} \left[\sin \left(\frac{n\pi}{2} t \right) \right]_{t=3}^{t=4} - \right. \\ &\quad \left. \frac{12}{n\pi} \left[t \sin \left(\frac{n\pi}{2} t \right) \right]_{t=3}^{t=4} - \frac{24}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{2} t \right) \right]_{t=3}^{t=4} \right\} \\ &= \frac{12}{n\pi} \sin \left(\frac{n\pi}{2} \right) - \frac{24}{n^2 \pi^2} - \frac{48}{n\pi} \sin \left(\frac{3n\pi}{2} \right) + \frac{36}{n\pi} \sin \left(\frac{3n\pi}{2} \right) - \frac{24}{n^2 \pi^2} \left[1 - \cos \left(\frac{3n\pi}{2} \right) \right] \\ &= \frac{12}{n\pi} \sin \left(\frac{n\pi}{2} \right) - \frac{12}{n\pi} \sin \left(\frac{3n\pi}{2} \right) - \frac{24}{n^2 \pi^2} \left[2 - \cos \left(\frac{3n\pi}{2} \right) \right] \\ &= \frac{24}{n\pi} \sin \left(\frac{n\pi}{2} \right) \cos(n\pi) - \frac{24}{n^2 \pi^2} \left[2 - \cos \left(\frac{3n\pi}{2} \right) \right] \end{aligned}$$

Step 5 of 5

Since the function is even, $b_n = 0$.

The Fourier series is,

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\ &= 3 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t) \\ &= 3 + \frac{24}{\pi} \sum_{n=1}^{\infty} \left\{ \left[\frac{1}{n} \sin \left(\frac{n\pi}{2} \right) \cos(n\pi) - \frac{1}{n^2} \left[2 - \cos \left(\frac{3n\pi}{2} \right) \right] \right] \cos \left(\frac{n\pi}{2} t \right) \right\} \end{aligned}$$

Thus, the Fourier series of the periodic function is,

$$f(t) = 3 + \frac{24}{\pi} \sum_{n=1}^{\infty} \left\{ \left[\frac{1}{n} \sin \left(\frac{n\pi}{2} \right) \cos(n\pi) - \frac{1}{n^2} \left[2 - \cos \left(\frac{3n\pi}{2} \right) \right] \right] \cos \left(\frac{n\pi}{2} t \right) \right\}.$$